Project Summary: Parkinson's Disease Classification using Voice Features

1. Dataset & Patient Recording Overview

For this project, I worked with the Parkinson's Disease Speech Features dataset. The dataset contains 756 voice recordings collected from 252 individuals, which means each patient provided exactly 3 separate recordings. My primary objective was to build a binary classification model that can accurately distinguish between Parkinson's Disease patients and healthy individuals based on their extracted vocal features.

2. Step-by-Step Methodology

Step 1: Data Splitting Strategies & Traditional Boosting Baseline

To establish a solid baseline, I started by evaluating three traditional boosting algorithms: XGBoost, LightGBM, and CatBoost. Since we have multiple recordings per patient, I tested two different ways to split our data to see how the models behave:

- Grouped Split (Patient Records Isolated): To simulate a strict, real-world diagnostic scenario, I ensured that all 3 recordings of a single patient were placed entirely into either the training set or the testing set.
- Random Split (Patient Records Mixed): I distributed the recordings randomly across the dataset. This meant different recordings from the same patient could appear in both the training and testing sets simultaneously.

Observation: I noticed that the Random Split strategy provided the highest predictive accuracies, with both LightGBM and XGBoost achieving 90.13%. Following our project's objectives, I selected this best-performing setup (LightGBM + Random Split) as the foundation for the subsequent deep learning experiments.

Step 2: Evaluating Deep Tabular & Lightweight Transformer Models

Using the chosen baseline data setup, I wanted to investigate whether advanced deep learning models designed specifically for tabular data could outperform the traditional boosting trees. I conducted experiments and carefully tuned the hyperparameters where applicable. The results were as follows:

- TabNet (Tuned): Achieved 86.84% accuracy.
- GatedTabTransformer (GATE): Achieved 85.53% accuracy.
- SAINT (FTTransformer): Achieved 84.21% accuracy.
- TabTransformer: Achieved 80.92% accuracy.
- NODE: Achieved 80.26% accuracy.

Step 3: Integrating Kolmogorov-Arnold Networks (KAN)

To explore a more novel neural architecture, I implemented the newly introduced Kolmogorov-Arnold Network (KAN). Interestingly, KAN significantly outperformed all the transformer-based models I tested, reaching a highly impressive 90.79% accuracy as a standalone deep learning model.

Step 4: Final Model Fusion Strategy

To build the ultimate predictive model for this project, I took the best traditional model (LightGBM) and our best-performing deep learning model (KAN). I combined their predictions using a Soft Voting Fusion technique, which averages the output probabilities of both models to make a final decision.

3. Final Results & Conclusion

The fusion strategy successfully combined the strengths of the tree-based boosting method and the neural network. This fusion yielded the highest overall accuracy in the entire study. More importantly, the fusion model achieved a 98% Recall for the patient class, meaning it successfully detected 98% of the actual

Parkinson's patients.

Comprehensive Model Accuracy Table

Methodology Category	Algorithm / Architecture	Accuracy
Traditional Boosting	LightGBM (Selected Best Baseline)	90.13%
Traditional Boosting	XGBoost	90.13%
Traditional Boosting	CatBoost	88.16%
Advanced Deep Learning	**Kolmogorov-Arnold Network (KAN)**	**90.79%**
Deep Tabular / Transformer	TabNet (Tuned)	86.84%
Deep Tabular / Transformer	GatedTabTransformer (GATE)	85.53%
Deep Tabular / Transformer	SAINT (FTTransformer)	84.21%
Deep Tabular / Transformer	TabTransformer	80.92%
Deep Tabular / Transformer	NODE	80.26%
Proposed Fusion Model	**LightGBM + KAN (Soft Voting)**	**91.45%**

Conclusion:

My systematic evaluation demonstrates that while patient-mixed random splitting inflates the initial boosting accuracy, integrating LightGBM with the modern KAN architecture via soft-voting fusion reliably provides the most optimal performance (91.45% Accuracy) for screening Parkinson's disease.